

Table 1: The 7Rs of migration

7Rs of Migration	Description
<i>Refactor</i>	Refactor or re-architect the application to take advantage of cloud-native features and capabilities.
<i>Replatform</i>	Upgrade or replace the underlying infrastructure of the application to take advantage of cloud services, without making significant changes to the application itself.
<i>Repurchase</i>	Move to a different software product or service, such as a SaaS offering like Office365, Google Workplace, ZohoOne, instead of migrating the application itself.
<i>Rehost</i>	Also known as “lift and shift,” migrate the application to the cloud with minimal changes to the application architecture, and use cloud infrastructure as a like-for-like replacement of on-premises infrastructure.
<i>Relocate</i>	Move the application to a different location or region, without making significant changes to the application itself.
<i>Retain</i>	Maintain the application on-premises or in a hosted environment, and do not migrate it to the cloud.
<i>Retire</i>	Decommission the application, either because it is no longer needed or because it will be replaced with a different application.

- **Online data transfer:** This method involves transferring data over the internet using a secure connection. It is typically the most convenient and cost-effective method for small to medium-sized data sets.
- **Offline data transfer:** For large data sets, offline data transfer may be more practical. This method involves physically transferring data to the cloud provider, typically using a secure storage device.
- **Hybrid data transfer:** In some cases, a combination of online and offline data transfer methods may be required to ensure efficient and secure data transfer. For example, a hybrid approach can be used for large healthcare organisations transferring patient data to the cloud while maintaining system uptime.

Validation of data accuracy and integrity: Validating the accuracy and integrity of transferred data is crucial to ensure a successful migration. Different methods can be employed for this. Thorough validation helps identify discrepancies or errors, and ensures the integrity of the data in the cloud environment.

Manual validation: This method involves manually comparing the data in the cloud environment with the original data source. It can be time-consuming, but is typically the most accurate method.

Automated validation: Automated tools can be used to compare the data in the cloud environment with the original data source. This method is typically faster than manual validation, but it may not be as accurate.

Sampling validation: In some cases, it may be impractical to validate all the data in the cloud environment. In these cases, a sampling of the data can be validated to ensure that it is accurate and consistent with the original data source.

Integration and interoperability

Integrating cloud services with on-premises applications and systems is critical to ensuring seamless operation. Proper integration can help businesses maximise the benefits of cloud migration and minimise disruption to operations.

Integrating cloud services with on-premises applications and systems: This is a critical step in the migrating process. It enables seamless access and use of data and applications between the cloud and on-premises environments. Organisations must identify the suitable integration patterns and protocols that align with their existing infrastructure. Examples include point-to-point integration, middleware-based integration, and API-based integration. Each pattern has its own benefits and drawbacks, and organisations need to choose the one that best fits their requirements. A hybrid integration platform can provide a unified solution. It typically includes a set of pre-built connectors, adapters, and APIs facilitating smooth integration between cloud services and on-premises systems.

Ensuring interoperability with third-party solutions: In addition to integrating on-premises systems, businesses must ensure interoperability with third-party solutions. These solutions can include software applications, platforms, and services that are not part of the organisation's IT infrastructure. Compatibility with third-party solutions is essential to leverage the benefits of the cloud and these external offerings.

Evaluation of integration patterns and protocols that are compatible with the third-party solutions, such as point-to-point integration, middleware-based integration, and API-based integration, is necessary. Organisations need to choose the pattern that best fits their requirements.